

Renewable energies and their role in achieving sustainable development

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ABSTRACT

Renewable energy is considered one of the most important available energy alternatives worldwide and a key element in achieving sustainable development. The relationship it utilizes to meet the requirements of sustainable development in Algeria, which has sought to invest in these resources, is through the establishment of a renewable energy and energy efficiency program. This program aims to enhance the reliance on renewable energy across various sectors to achieve economic and social objectives while preserving the environment.

Keywords: Renewable Energy, Sustainable Development, Energy Efficiency, Environment.

1 INTRODUCTION

The world is witnessing significant industrial development and continuous growth in the economies of major industrialized nations. In recent years, global energy demand has clearly risen. Amid this increasing need for conventional energy sources, the negative consequences on the environmental and developmental features of nations have intensified, so driving the quest for alternative energy sources to reach thorough and balanced sustainable development. This would allow the growth of national economies considering many facets of human existence.

Adopting plans and creating laws meant to find other energy sources has also become rather crucial. Moreover, initiatives for international cooperation have been investigated to promote the use of renewable energy sources in a manner that protects the environment and guarantees sustainable development in many sectors for several generations. Given that conventional fossil energy sources are depletable and now the primary contributors to the rising emissions of hazardous gases into the atmosphere, which generate global warming, this is especially crucial.

As for Algeria, the development of renewable energy is of great importance to the public authorities, who aim to replace fossil fuels with renewable energy across various sectors. This is to achieve specific objectives in line with the plans and strategies followed in this context, particularly the goal of energy independence from oil and meeting the requirements of sustainable development. To this end, the government has established an ambitious program to develop renewable energy, improve energy efficiency, and implement a suitable legislative and regulatory framework.

From here, we raise the following question: What role do renewable energies play in achieving sustainable development in Algeria?

2 THE CONCEPT OF RENEWABLE ENERGIES AND THEIR IMPORTANCE

2.1 DEFINITION OF ENERGY

Energy is the ability of matter to exert a force capable of accomplishing a specific task. It is the capacity of a system to produce effectiveness or external activity. It is a physical quantity that manifests in the form of heat, mechanical motion, or as binding energy in atomic nuclei or protons and neutrons (Abdullah 2016, 160).

2.2 DEFINITION OF RENEWABLE ENERGY

Renewable energy refers to energy that recurs naturally and periodically in nature. It is energy generated from an inexhaustible natural source, readily available across the Earth's surface, and can be easily converted into usable energy. Renewable energy is characterized by being environmentally friendly, clean, and non-polluting. Unlike non-renewable energy, which is depletable, the use of renewable energy does not lead to its exhaustion. In contrast to conventional energy, such as solar and wind energy, which require transportation mechanisms that are present in most countries, renewable energy is a local source that remains consistent with the reality of development. Development depends on three key factors: economic, social (improving infrastructure and ensuring human rights), and environmental (balancing nature with humanity and preserving the ecological system). Renewable energy ensures continuous availability at an affordable price, does not produce noise, and leaves no harmful waste that could damage the environment. (Abdullah 2016, 160).

2.3 IMPORTANCE OF RENEWABLE ENERGY

2.3.1 Energy Security

Although the concept of energy security is relatively modern and has gained notable attention in recent years, the pursuit of securing energy needs by societies and nations is not a recent phenomenon. Attempts to control energy sources were a fundamental driving force behind many past conflicts, and were also a factor in fostering rapprochement and cooperation between various countries. In fact, the term "energy security" was directly used during World War I. With the continuous increase in global energy consumption, the reserves of fossil energy sources, particularly petroleum, are dwindling, leading to their depletion and the potential for exhaustion. This has prompted the consideration of alternative sources to sustain energy, which has indeed been the case, as statistics indicate (Rawaikiya 2019, 107).

Global consumption of renewable energy has consistently increased, reaching approximately 316.9 million tons of oil equivalent (Mtoe) in 2014 compared to 283 Mtoe in 2013, thus accounting for 2.5% of global energy consumption (Rawaikiya 2019, 107).

2.3.2 Climate Change

The use of renewable energy will reduce the emissions of greenhouse gases.

2.3.3 Decreasing Cost of Renewable Energy

The decreasing cost of renewable energy is one of the key incentives driving countries worldwide to adopt renewable energy sources. The early years of interest in these sources saw an increase in costs, but this has steadily decreased, primarily due to improvements in the technology used to produce these energies. This decline will continue as the industry matures.

2.3.4 The Necessity of Diversifying Energy Sources

To reduce the pressure on the use of petroleum as a depleting resource, it is essential to rationalize the use of these resources and find new, renewable energy alternatives.

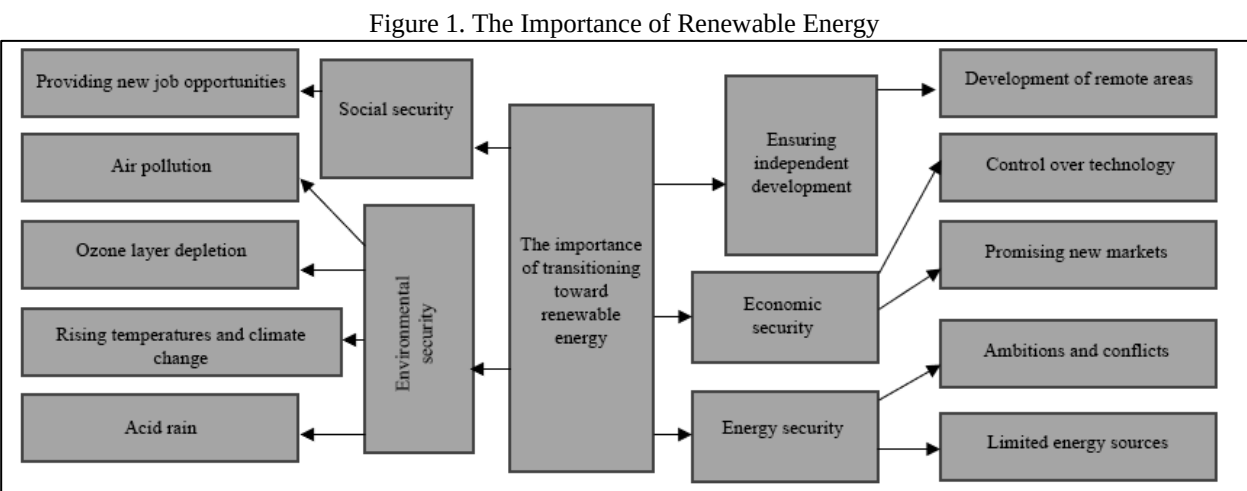
2.3.5 The Abundance of Studies on Alternative Energies

Numerous conferences have been held on this topic. For example, in 2001, an international conference on renewable energy was organized, during which the urgent need to accelerate its exploitation was emphasized, given the ever-increasing demand for energy over time.

Given that oil-producing countries, in particular, hold significant importance in the global energy sector, their shift toward developing renewable energy use will offer them many other advantages, including:

- The shift of oil-producing countries towards renewable energy can help preserve their depleting resources.
- Investment by oil-producing countries in renewable energy will transform them from oil and gas producers and exporters into important international players in the energy sector.
- The diversification of energy sources in oil-producing countries enables them to achieve energy security.
- By diversifying energy sources, oil-producing countries can preserve their resources and extend the period of their exploitation.
- Diversification of energy sources in oil-producing countries enhances and develops their human capital.

The following figure summarizes the importance of renewable energy in terms of securing sustainable development, food security, and energy security itself (Rawaikiye 2019, 108).



Source: (Tekwante 2012).

3 TYPES OF RENEWABLE ENERGY

Renewable energy takes various forms and types based on the nature and source of its origin. The most prominent of these include solar energy, wind energy, tidal energy, geothermal energy, hydroelectric energy, biofuels, and biomass energy.

3.1 SOLAR ENERGY

The sun is considered the essential source of energy necessary for life on Earth, and it is the primary source of energy in all its forms, whether fossil or renewable. Solar energy is the most important among new energy sources, with many countries investing substantial efforts through scientific research to develop methods for harnessing it as an alternative to oil and gas (Kamal and others 2018, 06).

Solar energy is considered the largest alternative to petroleum in electricity generation. It is a relatively new technology and holds a promising position in the energy field. Its resources are vast, its environmental impact is limited, and it provides countries that are most exposed to sunlight with an opportunity similar to that which is currently provided by offshore wind farms in European countries with wind-exposed coastlines (Laajal 2020, 165).

3.2 WIND ENERGY

Wind energy is converted into electrical energy using large turbines. It is the fastest-growing and most rapidly expanding renewable energy source worldwide. Germany leads the world in wind energy utilization. Despite the widespread expansion of wind farms, they face some environmental challenges, particularly regarding the large areas of land they require (Laajal 2020, 165).

3.3 TIDAL ENERGY

Tidal energy is a type of mechanical energy stored in the currents generated by the rising and falling tides, which are naturally caused by the gravitational pull of the moon and the sun, as well as the Earth's rotation around its axis. This energy is classified as renewable, and these two phenomena are exploited by utilizing the currents stored in water during the occurrence of the tidal cycles.

3.4 GEOTHERMAL ENERGY

Geothermal energy is a clean and renewable energy source. It is high-temperature thermal energy of natural origin, stored within the Earth's interior. This thermal energy is primarily used for electricity generation, and the heat is extracted from underground reservoirs through wells or other methods (Al-Taif 2019, 171).

3.5 HYDROELECTRIC ENERGY

Hydroelectric energy is the electrical power generated from the potential energy of water through the utilization of its movement. This type of energy harnesses the power of water motion to generate electricity and is considered to be extremely clean with widespread use (Al-Taif 2019, 171).

3.6 BIOFUEL ENERGY

Biofuel is one of the most important forms of renewable energy, serving as a suitable alternative to oil. It contributes to providing healthy supplies derived from alternative energy sources. Some of the most commonly used plants in sustainable biofuel production include sugarcane and algae.

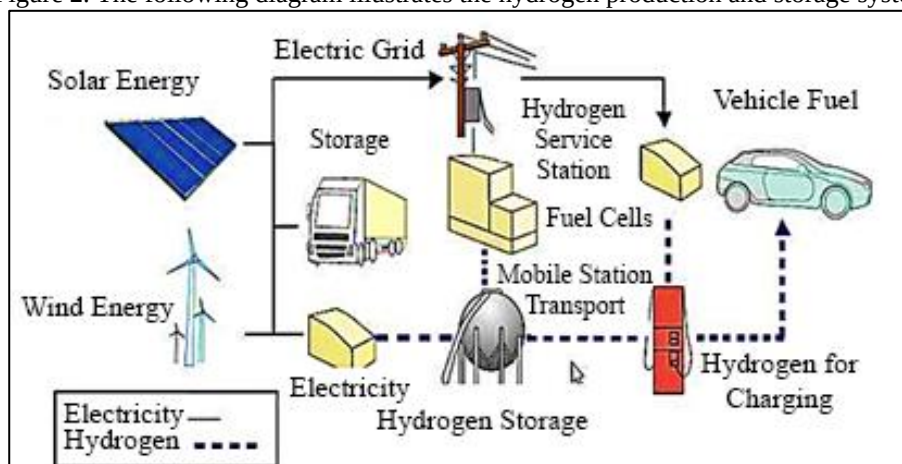
3.7 BIOMASS ENERGY

Biomass energy refers to the energy derived and extracted from all types of materials related to living organisms, such as wood, grasses, and animal waste. A drawback of this energy is that the combustion of these organic materials results in the emission of carbon dioxide (Al-Taif 2019, 171).

3.8 GREEN HYDROGEN ENERGY

Green hydrogen is produced through the electrolysis of water in an electrolyzer, with electricity supplied from renewable energy sources such as hydroelectric power, wind energy, or solar energy. If desalinated water is required, the rest of the process is fully supported by solar and wind energy, ensuring that no greenhouse gases are emitted during this process. (Badri Abdel Aziz, 2018, 2019, p. 69).

Figure 2. The following diagram illustrates the hydrogen production and storage system



Source:(Saadi 2016, 02).

4 DEFINITION OF SUSTAINABLE DEVELOPMENT

Sustainable development has several definitions, including the following: Sustainable development refers to the management and use of natural resources, such as soil and water, in an efficient manner without waste or delay, along with continuous maintenance. It also involves the rehabilitation of the investment and utilization processes for these resources, coupled with human development in various forms, whether economic, health-related, educational, recreational, or cultural. All of this aims to meet both current and future human needs through a network of responsible institutions and in accordance with multiple techniques, as well as the socially acceptable modern economic and technological capabilities. (Al-Samarrai 2016, 09).

The definition from the Brundtland Commission in 1987, in the report *Our Common Future*, is the most famous and widely accepted in the scientific literature. It emphasizes the protection of future generations' resources. According to the Commission, sustainability means:

Meeting the needs of the present generation without compromising the ability of future generations to meet their own needs. It should be noted that the essential idea behind this definition lies in the "principle of the necessity to preserve resources for future generations (Aouali 2018, 169)."

5 CHARACTERISTICS OF SUSTAINABLE DEVELOPMENT

Sustainable development has several characteristics, as follows:

- Sustainable development differs from general development in that it is more deeply interconnected and more complex, especially concerning what is natural and social in development.

- Sustainable development means bringing about changes in all aspects of economic life, represented by an increase in the per capita real income as well as preserving natural resources, whether renewable or non-renewable, through their rational exploitation.
- Sustainable development is a continuous process, both in the present and the future, which satisfies the needs and aspirations of both the present and future generations. The state aims to achieve development across all sectors to benefit future generations.
- Sustainable development is an inclusive and shared responsibility, encompassing all state sectors, and it is the duty of the state at all levels to contribute to the decision-making process.
- The term "sustainable development" is considered a global term, derived from political, economic, and cultural studies that have contributed to introducing a concept that embodies sustainable development.
- Sustainable development has goals it seeks to achieve through effective mechanisms and principles upon which it is based (Khaled 2020, 61).

6 DIMENSIONS OF SUSTAINABLE DEVELOPMENT

The dimensions of sustainable development, based on the previous definitions, encompass a set of interrelated aspects that aim to achieve sustainable development. These dimensions are economic, human, environmental, and technological.

6.1 ECONOMIC DIMENSIONS

Through focusing on:

- Determining the share of individual consumption from natural resources
- Stopping the waste of natural resources
- Addressing the pollution issue in developed countries and finding solutions
- Reducing the dependence of developing countries
- Focusing on sustainable development in poor countries
- Ensuring equality in resource distribution
- Reducing income disparities
- Cutting down military expenditures

6.2 HUMAN DIMENSIONS

- Determining the final population size
- The importance of population distribution
- Full utilization of human resources
- Health and education
- Highlighting the significance of women's roles
- Achieving a democratic socialist system of governance

6.3 ENVIRONMENTAL DIMENSIONS

- Soil degradation, pesticide use, destruction of vegetation, and fishing grounds
- Protection of natural resources
- Water conservation
- Reduction of habitats for biological species
- Protecting the climate from global warming

6.4 TECHNOLOGICAL DIMENSIONS

- Use of cleaner technologies in industrial facilities
- Adoption of advanced technologies and regulatory legal frameworks
- Fuels and global warming
- Reducing gas emissions
- Preventing the degradation of the ozone layer

7 SUSTAINABLE DEVELOPMENT GOALS

The Sustainable Development Goals (SDGs) consist of a diverse set of objectives that cover various aspects influencing human life and the entire planet. These goals include areas such as the eradication of poverty in all its forms, achieving gender equality, ensuring quality education, guaranteeing access to sanitary facilities and clean water, combating climate change, and safeguarding the environment, and promoting partnerships to achieve these objectives, among many others.

All United Nations member countries in 2015 approved the SDGs meant to eliminate poverty, lower inequality, and create more peaceful and prosperous societies by 2030. A call to action for building a world where no one is left behind, the SDGs—also called the "Global Goals".

More than 100 UN member states have committed to children's rights in the context of implementing the SDGs. UNICEF works with governments, partners, and other UN agencies to assist countries in ensuring that these goals result in tangible outcomes for every child, both for today's generations and future ones.

- Eradication of poverty in all its forms - Complete eradication of hunger
- Gender equality, high-quality education, and good health and well-being
- Clean water and sanitation – Affordable and clean energy
- Decent work and economic growth – Industry, innovation, and infrastructure
- Reduced inequalities – Sustainable cities and communities
- Responsible consumption and production – Climate action
- Life below water – Life on land
- Peace, justice, and strong institutions – Strengthening the means of implementation and revitalizing the global partnership for sustainable development (Aboud 2023, 496)

Figure 3. illustrates the Sustainable Development Goals



Source: (United Nations Educational, Scientific and Cultural Organization n.d.).

8 SUSTAINABLE DEVELOPMENT AND ITS RELATION TO RENEWABLE ENERGIES

The direction of sustainable development in dealing with natural resources in a sustainable manner means considering the limited nature of resources and avoiding environmental pollution. Therefore, renewable energies represent hope and the future for energy production, replacing traditional sources. The United Nations has held numerous conferences to achieve this goal.

8.1 SOCIAL AND ECONOMIC DIMENSIONS OF SUSTAINABLE DEVELOPMENT RELATED TO THE ENERGY SECTOR

The United Nations Conference on Environment and Development, held in June 1992, marked the beginning of an important phase of international cooperation founded on acknowledging the need to combine environmental issues and development. This integration guarantees the sustainability of the development process, which requires meeting the economic and social development needs of current generations while preserving the environment and considering the rights of future generations to development. The chapters in the first part titled "Social and Economic Dimensions of Sustainable Development" of Agenda 21 include goals and activities related to the energy sector, as follows:

8.2 IN THE FIELD OF COMBATING POVERTY

Providing sustainable living conditions for populations in impoverished areas, relying on the sustainable management of available resources, while also offering appropriate job opportunities and favorable health and educational conditions to enhance economic growth in these regions.

8.3 PROPOSED ACTIVITIES INCLUDE

- Developing appropriate infrastructure for development and suitable technologies for use in these areas.
- Introducing the necessary standards for managing and preserving available resources to meet the basic needs of the population.

8.4 IN THE FIELD OF CHANGING CONSUMPTION PATTERNS

Encouraging sound production and consumption patterns that meet human basic needs while focusing on environmental preservation; achieving production efficiency and changes in consumption patterns to focus on optimizing resource use and minimizing waste. This requires a shift in the attitudes of governments, families, and individuals through the following:

- Promoting increased efficiency in the use of available energy and resources;
- Minimizing emissions to the lowest possible level to reduce pollution;
- Helping individuals and families make environmentally sound purchasing decisions;
- Implementing environmentally sustainable energy pricing policies, taking into account the social conditions of poor populations, and supporting values that promote responsible consumption;
- Supporting research and development activities in the field of consumption assessment, as well as introducing new concepts for resource use in ways that foster successful economic growth and the development of regional and international cooperation programs aimed at achieving sustainable consumption patterns.

8.5 IN THE FIELD OF SUSTAINABLE DEVELOPMENT OF HUMAN SETTLEMENTS

This section includes two energy-related programs:

- "Achieving Integration in the Provision of Basic Services in Locations" Program: Its activities involve providing settlement areas with the necessary infrastructure, including energy and water facilities, by 2025. All developing countries should incorporate activities related to technical, financial, and human resources into their national programs and strategies to improve the integration between infrastructure and environmental planning.

8.6 PROGRAM FOR PROVIDING SUSTAINABLE ENERGY AND TRANSPORT SYSTEMS IN HUMAN SETTLEMENTS

This program includes several activities to achieve sustainable development in settlements, such as:

Providing the necessary technologies and practical solutions to improve energy efficiency: Renewable energy technologies have been developed, and their applications have been expanded to serve human settlements, reducing the harmful environmental impacts of energy production and consumption on human health, particularly those related to air pollution and water sources in urban areas.

9 THE ROLE OF RENEWABLE ENERGY IN ACHIEVING SUSTAINABLE DEVELOPMENT IN ALGERIA

Algeria possesses significant potential in the field of renewable energy, particularly solar energy. Renewable energy is considered a key driver for sustainable economic development and is central to national energy and economic policies through 2030. It is expected to provide approximately 40% of energy needs, primarily from photovoltaic and thermal solar energy.

9.1 RENEWABLE ENERGY AND ENERGY EFFICIENCY PROGRAM

In 2011, Algeria launched an important program for the development of renewable energy, extending over the next two decades. In addition to the environmental aspect and the commitment to diversifying energy sources, Algeria also aims to determine the lifespan of its hydrocarbon reserves and exploit its vast solar and wind energy potential. This effort aims to meet internal electricity needs and export a portion of this energy to European countries.

9.2 RENEWABLE ENERGY PROGRAM

The renewable energy projects aimed at electricity production for the national market are divided into three phases (Kloum and Iman 2011, 05-07).

The program, from now until 2020, includes the implementation of sixty (60) solar thermal plants, wind energy fields, and various other stations. The first phase (2011-2013) focuses on the realization of pilot (model) projects and testing various available technologies. The second phase (2014-2015) involves direct changes in program deployment. The third phase (2016-2020) is dedicated to large-scale implementation.

9.3 ENERGY EFFICIENCY PROGRAM

The energy efficiency initiative tackles Algeria's wish to encourage the best and responsible use of energy sources. Energy efficiency aims to generate the same advantages or services with as little energy as possible. Among the components of the energy efficiency program are:

- The use of thermal insulation in buildings to lower energy use connected to heating and cooling.
- The installation of solar water heaters to replace traditional water heaters
- The widespread use of low-energy consumption light bulbs

- Encouraging local production by creating partnerships between local and foreign producers (Ministry of Energy and Mines 2011, 05)

10 CONCLUSION

The significance of the role that renewable energies play in achieving the dimensions of sustainable development, while preserving the environmental heritage for future generations, highlights the necessity for countries to adopt a development model based on renewable energies. Given that Algeria is rich in renewable energy resources, particularly solar energy, it is well-positioned to become a leading country in the export of clean energy. For this reason, the Algerian government has paid considerable attention to the development and utilization of renewable energies, considering the available resources.

To achieve this, Algeria has established numerous bodies and centers dedicated to this cause and has developed long-term strategies and programs to promote renewable energy. Furthermore, significant financial resources have been allocated to encourage investment in this sector, with some initiatives already implemented and others currently underway.

Although its energy production from these sources remains below the expected level and available potential, it seeks to harness these sources for its sustainable development.

This can only be achieved by creating a conducive environment that encourages investment in this area, as well as providing full government support for both the preservation of renewable energies and the various related centers. This will enable the country to achieve the objectives for which they were established.

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